

Assignment 3 Report

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1 Introduction

In this report, I explore the challenge of estimating physical constants—temperature (T), speed of light (c), Planck's constant (h), and Boltzmann constant (k)—from Planck's formula using experimental spectral radiance data. This task is crucial for understanding radiative energy, and my experience with this problem highlighted the nuances of parameter estimation in complex models.

2 Planck's Law of Spectral Radiance

Planck's formula for spectral radiance $B(\lambda, T)$ is given by:

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5 (\exp(\frac{hc}{\lambda kT}) - 1)}$$

where:

- $B(\lambda, T)$: Spectral radiance
- h : Planck's constant
- c : Speed of light
- k : Boltzmann constant
- λ : Wavelength
- T : Temperature

3 Challenges in Fitting All Four Parameters Simultaneously

I encountered significant challenges when trying to fit all four parameters (T , c , h , k) simultaneously:

- The optimization turned out to be highly non-linear with numerous local minima.
- Convergence issues were frequent. I found that the fitting process could easily diverge due to the strong interdependencies among parameters.
- Minor adjustments in one parameter had substantial effects on others, complicating the overall fitting process and leading to inaccurate results.

These experiences highlighted the complexity of multi-parameter optimization and the need for more targeted approaches.

4 Partial Application and Single Parameter Fitting

To address these challenges, I used partial application, fixing some parameters to their known values:

- This method simplified the problem by reducing its dimensionality. For instance, by fixing c , h , and k , I found that estimating T became more manageable and yielded more accurate results.
- This approach allowed me to isolate the effect of a single parameter, which made the optimization process more focused and less prone to the issues seen in full parameter fitting.

5 Pair Estimation for Improved Accuracy

Pair estimation emerged as a middle ground between full parameter fitting and single parameter fitting:

- It allowed simultaneous adjustment of two interdependent parameters, capturing their combined influence more effectively.
- For example, estimating T and h together provided insights into how temperature and Planck's constant interact, leading to more realistic parameter values.
- I found that pair estimation often struck a better balance, addressing interdependencies without the excessive complexity of fitting all four parameters.

6 Comparing Estimation Approaches

Based on my experiments:

- **Full Parameter Estimation:** It often led to convergence issues and inaccuracies due to the complex parameter interactions.
- **Single Parameter Estimation:** Simplified the problem but occasionally missed important interactions between parameters.
- **Pair Estimation:** Offered a practical balance, improving fitting accuracy by accounting for parameter interdependencies without the full complexity of simultaneous fitting.

7 Results and Observations

The estimated parameters obtained through various fitting models are as follows:

7.1 Single Parameter Estimation

For single parameter estimation, the following values were obtained:

- **Model T_c:** Temperature (T) = $3.94\text{e}+03$ K, Speed of light (c) = $2.92\text{e}+08$ m/s
- **Model T_h:** Temperature (T) = $3.83\text{e}+03$ K, Planck's constant (h) = $6.28\text{e}-34$ J·s
- **Model T_k:** Temperature (T) = $6.18\text{e}+03$ K, Boltzmann constant (k) = $8.94\text{e}-24$ J/K
- **Model c_h:** Speed of light (c) = $2.30\text{e}+08$ m/s, Planck's constant (h) = $1.07\text{e}-33$ J·s
- **Model c_k:** Speed of light (c) = $2.92\text{e}+08$ m/s, Boltzmann constant (k) = $1.09\text{e}-23$ J/K
- **Model h_k:** Planck's constant (h) = $6.28\text{e}-34$ J·s, Boltzmann constant (k) = $1.06\text{e}-23$ J/K

7.2 Pairwise Parameter Estimation

For pairwise estimation, the fitted parameters were:

- **Fitted Temperature (T):** $4.00\text{e}+03$ K
- **Fitted Speed of Light (c):** $4.32\text{e}+08$ m/s
- **Fitted Planck's Constant (h):** $8.75\text{e}-34$ J·s
- **Fitted Boltzmann Constant (k):** $1.11\text{e}-23$ J/K

These observations highlight the sensitivity of the fitting process to initial guesses and parameter interdependencies. Pairwise estimation provided more reliable values compared to single parameter fitting and was more manageable than full parameter fitting.

8 Conclusion

In conclusion, the fitting process for Planck's law parameters demonstrated the challenges associated with multi-parameter optimization. While full parameter fitting was comprehensive, it was complex and prone to convergence issues. Single parameter fitting simplified the process but sometimes overlooked interactions. Pair estimation emerged as a practical approach, providing a good balance between accuracy and complexity. This approach proved effective in obtaining realistic estimates for the physical constants involved.